Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Monitoring systems such as spacecraft and industrial plants must flag sensor faults quickly while staying quiet during normal operation. Detectors that reconstruct raw signals fire often on harmless jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) instead predict compact learned summaries and raise fewer false alarms, yet single-score JEPAs sum every coordinate into one number, letting drifting coordinates inflate the threshold. We replace that scalar score with per-coordinate circuit breakers: each coordinate's error across future horizons receives its own healthy-calibrated threshold, and an alarm fires when a quorum trips, with the quorum fitted to the correlated trip counts seen on healthy data. Laminar-JEPA-Lite, a 128-breaker design, is the operational detector; Laminar-JEPA, a 224-breaker variant, is kept as an ablation reference. On 118 untouched archive streams from nine datasets, with model fitting, checkpoint selection, adaptation, and calibration on disjoint healthy blocks, Laminar-JEPA-Lite raises live event recall from 52.5 to 63.8 percent over a single-score multi-horizon JEPA on every dataset and nearly halves detection delay. Pointwise F1 rises only modestly overall, from 0.076 to 0.079, but doubles when at least 500 calibration windows are available. These gains come mainly from the label-free operating point; at matched alarm budgets the panel performs on par with simpler coordinate pooling. False alarms have a low median of 0.78 percent but a heavy tail, with a 95th percentile of 50 percent caused mostly by healthy regime shifts, so autonomous use requires adequate calibration, a held-out acceptance check, and drift monitoring. Slow ramps remain undetected.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Spacecraft, power plants, and industrial machines stream readings from dozens of sensors at once [1]–[4]. A detector watching this telemetry has two jobs: raise an alarm soon after something breaks, and stay quiet the rest of the time. Faults are rare and seldom labeled, so the detector must learn what normal looks like from healthy history alone. Operators lose trust quickly when a monitor alarms several times a day on healthy data, and an ignored monitor catches nothing.

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The most common approach trains a network to reproduce raw sensor signals and raises an alarm when the reproduction is poor; TimesNet [5], TranAD [6], and Anomaly Transformer [7] follow this recipe. In practice, much of raw telemetry cannot be copied at all. Random sensor jitter leaves an irreducible error, and benign thermal drift or duty-cycle changes degrade the copy even when nothing is broken. Summing errors across 25–55 channels measured in different units allows one noisy channel to dominate, producing heavy-tailed spikes on healthy data.

These spikes defeat threshold calibration. Setting thresholds on healthy windows with Streaming Peaks-Over-Threshold (SPOT) [8] still yields 22%–24% causal false-positive rates (FPR) for reconstruction detectors on later healthy data across our 19 design-stage development streams, roughly one step in four. Because faults span many steps, alarming at random 24% of the time already hits 60.0% of fault episodes; TimesNet catches 58.8%, reflecting frequent alarms rather than selective fault recognition.

A Joint-Embedding Predictive Architecture (JEPA) [9], [10] avoids copying raw signals. An encoder compresses each window into a 32-number summary, and a predictor forecasts the summary of the next window. Discarding unpredictable jitter yields lighter error tails and cuts healthy false alarms on those development streams by an order of magnitude (Contribution 3).

Standard JEPA detectors, however, still collapse the 32 prediction errors into one scalar score and one threshold. When one or two summary features drift slightly as a machine warms up or switches load, their growing errors push the shared threshold upward. A fault that shifts the remaining 30 features moderately then stays below that raised threshold and goes missed. We call these drifting features *hijacker dials*. On our development streams, single-score JEPAs reach causal Point-F1 (step-level alarm accuracy) of only 0.050–0.055 and need 138–154 steps to flag a fault.

Two evaluation shortcuts also inflate published numbers: point adjustment credits an entire fault segment if one step inside it is flagged [11], and thresholds tuned with test labels or calibrated on adaptation data understate false alarms. We score every step without point adjustment, fit thresholds on healthy data only, separate model fitting, checkpoint selection, adaptation, and calibration into disjoint healthy blocks on our primary archive, and reserve 118 streams never inspected during design.

We replace the single scalar alarm with a panel of circuit

breakers, like a building breaker box where a short in one circuit trips its own switch regardless of total load. Across target prefixes of 16, 32, and 64 steps, each of the 32 coordinates receives separate *dials* tracking horizon error size and persistent drift, each with its own healthy-calibrated threshold, and the system alarms when a quorum of dials trips together. Drifting dials widen only their own thresholds instead of blinding the rest of the panel. Unlike a bank of independent detectors, the dials are structured views of one error matrix (per-horizon error size and persistent cross-horizon displacement), and the quorum is fitted to the joint trip counts seen on healthy data, so dial correlation is measured, not assumed away. Targeting abrupt to medium-onset faults (within roughly one 256-step context window), we study two variants (Fig. 1):

1) *Laminar-JEPA-Lite*, our preferred operational architecture, predicts three horizons in parallel behind one layer-normalization (LayerNorm) step [12] and monitors $4 \times 32 = 128$ position-plus-drift dials under one quorum (35%–52% fewer parameters and $2.43\times$ faster inference).
2) *Laminar-JEPA*, a 224-dial ablation reference variant, scores four views of each coordinate's error plus 32 onset relays and optional weight editing to isolate which views aid live causal versus buffered attribution; more dials do not imply a better operational detector.

A. Contributions

Thirty-three development streams (where adaptation and thresholds share calibration data) provide exploratory design-stage evidence, while 118 untouched archive streams across 9 datasets (of 141 streams across 12 datasets) under a disjoint five-block protocol provide primary evidence of generalization and calibration (Section V-B). We report live *causal* streaming separately from *buffered* after-the-fact attribution:

1) *Live event coverage and delay:* On the 118 untouched streams, *Laminar-JEPA-Lite* raises causal event recall (ER) from 52.5% to 63.8% over a single-score multi-horizon JEPA on all 9 datasets (sign-flip $p = 0.0039$; all 12 leave-one-dataset-out (LODO) folds) and nearly halves detection delay (70.8 vs. 133.5 steps). Pooled causal Point-F1 rises only modestly ($0.076 \rightarrow 0.079$; 95% stream confidence interval (CI) $[-0.019, +0.023]$) but doubles on streams with $N_{\text{cal}} \geq 500$ calibration windows ($0.040 \rightarrow 0.079$, $p = 2.7 \times 10^{-5}$), where the scalar baseline is otherwise preferable on Point-F1.
2) *Operating point and false-alarm audit:* At matched healthy alarm rates, the panel performs on par with Fisher, top-k, and max-z pooling of the same residuals, and at matched test FPR it shows no ranking advantage over the scalar baseline, so its gains come mainly from where its label-free quorum sets the operating point. Causal held-out FPR has median 0.78% but 95th percentile 50.2%; a stream-level audit traces this tail mainly to healthy regime shifts rather than to calibration length alone, and we state explicit conditions for autonomous deployment.
3) *Design-stage comparison with reconstructors:* On 19 development streams, latent prediction holds causal FPR at 0.6%–2.4% versus 22.2%–24.3% for reconstructors, and at a matched 1% test FPR *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events versus 21.7% for TimesNet.
4) *Explicit limits:* Proposition 1 gives geometric bounds whose scalar score reweighting is null; most streams have too few effective calibration samples to estimate a 99.5% quorum quantile precisely; an optional channel branch lifts recall on synthetic single-sensor faults (80.7%–96.5%) only by raising test FPR from 1.5% to 6.8%–19.4%; LayerNorm and centering effects are design-stage findings; and slow ramps ($L \geq 4C$, including all GHL streams) and natural sensor failures remain out of reach.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [14]. Contrastive approaches—temporal-contextual contrasting [15], TS2Vec [16], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [17]—depend on whether synthetic perturbations or patch discrepancies match test faults (Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [20] or variance-covariance terms [21]–[23].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], Hamiltonian/Lagrangian nets [27], [28], and Koopman lifts [29]–[33] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [34]; we instead borrow the algebraic mean-plus-fluctuation identity of Reynolds averaging [35] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [36] and Multivariate Exponentially Weighted Moving Average (MEWMA) [37] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [38] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [39]. Per-coordinate alternatives such as univariate Cumulative Sum (CUSUM) banks [40] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate

(FDR) control [41] hit a discrete calibration floor $(N_{\text{cal}}+1)^{-1}$ on short histories (Section VI-A). All methods are evaluated without point adjustment [11], [13], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Each test case in this paper is a *stream*: one recording from one system, with K sensor channels sampled over time. Every stream has a healthy (nominal) history, used for training and calibration, followed by a labeled test segment that contains the faults. We use two collections of streams. A 45-stream development suite drawn from 11 datasets guided every design decision and supplies the main development benchmark. A larger 141-stream archive drawn from 12 datasets supplies the final evaluation, and 118 of its streams were never inspected during design.

Development suite: The 45 streams come from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45]. They cover three kinds of systems with channel counts from 2 to 72 and very different fault types:

1) *Spacecraft and solar telemetry:* six SMAP streams (K=25), six MSL rover streams (K=55), and SWAN-SF solar magnetograms [46] (K=38). NASA labels each SMAP and MSL fault as either an out-of-range *point* fault, where a value leaves its normal range, or an in-range *contextual* fault, where values stay in range but behave abnormally together [4] (Table V(a)).
2) *Wearable motion sensing:* six Daphnet Freezing-of-Gait (FoG) streams [43] recorded at 64 Hz (K=9, 72 episodes), so the S=64-step target window spans one second.
3) *Industrial processes and infrastructure:* 14 Gasoil Heating Loop (GHL) streams (K=16) whose 26 faults are slow ramps, the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72).

Thirty-three of these streams form the development benchmark: 19 non-GHL streams and the 14 GHL streams, spanning 7 datasets. Inclusion required enough healthy data for calibration ($N_{\text{cal}} \geq 500$ windows), at most 55 channels, and at most 1.2×10^5 test steps (Table IV(c)). The other 12 streams are reserved: five have very short healthy histories (764–1,492 steps; CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), the six CICIDS streams exceed the channel limit, and GECCO is evaluated in the archive instead.

Evaluation archive: The archive contains every qualifying stream from 12 datasets, requiring at least 100 calibration windows and at least one labeled anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, and 13 OPPORTUNITY streams, plus single streams from GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN. Healthy histories range from 1,920 to 132,481 steps and test segments from 2,158 to 114,001 steps. Removing the 23 streams used to tune the method leaves 118 untouched streams from 9 datasets (Section V-B).

How each stream is split: Every healthy history is divided in time order. The first 65% fits the model, and its first part also sets the per-channel standardization ($\boldsymbol{\sigma}$ floored at 10^{-2}). The next 17.5% calibrates alarm thresholds ($\mathbf{X}_{\text{cal}}$), and after a 64-step gap the last 17.5% measures false alarms on healthy data the thresholds have never seen ($\mathbf{X}_{\text{hold}}$). Archive runs further divide the model-fitting block (Section V-A).

Caveats: SMAP and MSL contain documented labeling flaws [42], and streams from the same archive often share hardware, so they are not fully independent. We therefore also treat whole datasets as units of analysis (7 for the development benchmark, 9 for the untouched streams, 12 for the full archive), using Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out audits (Section V-A).

IV. Methodology: Laminar-JEPA and Laminar-JEPA-Lite

Fig. 1 shows the pipeline from raw telemetry $\mathbf{X} \in \mathbb{R}^{T\times K}$ (K channels over T steps) to alarms. In plain terms, it has four stages: (1) encode each window of recent data into a short summary vector; (2) predict the summaries of the next few stretches of data; (3) turn the prediction errors into a set of *dials*, each measuring one aspect of the error; and (4) give every dial its own threshold and raise an alarm when enough dials trip together. The networks are trained on healthy data only; stabilization weights and alarm thresholds are then fitted on later healthy blocks and frozen.

A. Stage 1: Multiscale Encoder and Barycenter Centering

A sliding window of length $W = C + S = 320$ advances one step at a time. At each position t it splits into a *context window* $\mathbf{x}_{\text{ctx}}$ of the past $C = 256$ steps and a *target window* $\mathbf{x}_{\text{tgt}}$ of the next $S = 64$ steps, which ends at the decision time $\tau = t+S-1$. Telemetry mixes slow trends with fast vibration, so the encoder first separates time scales. Multichannel inputs ($K > 1$) pass through a causal per-channel convolution and are split into four smooth frequency bands (dyadic shells of a 256-point real fast Fourier transform (rFFT), forming an amplitude partition of unity). Each band has its own causal convolution (kernels $\{15, 9, 5, 3\}$), and a softmax router feeds the bands into a causal dilated temporal convolutional network (TCN) [48]. Single-channel inputs skip the band split.

The encoders follow the Joint-Embedding Predictive Architecture (JEPA) [9]: instead of reconstructing raw waveforms, an online encoder E_θ and a target encoder E_ϕ map context and target windows to $D = 32$-dimensional summaries, each normalized by LayerNorm to length close to $\sqrt{D}$. The target encoder is a slowly updated copy of the online one, an exponential moving average (EMA) with momentum $m = 0.996$ (0.995 on archive cohorts) [18], [20]. Two problems arise in practice. Deep dilated TCNs blur brief spikes at fault onset, and changes in operating regime shift all summaries together. Two encoder components address them:

1) *Multi-depth tapping (keeps fast transients):* Each encoder also reads three intermediate feature maps ($d_{\text{model}} = 48$): the spectral stem, a middle TCN block, and the final TCN block. For each tap, the last time step and the root-mean-square (RMS) energy over time are projected to $\mathbb{R}^D$ by

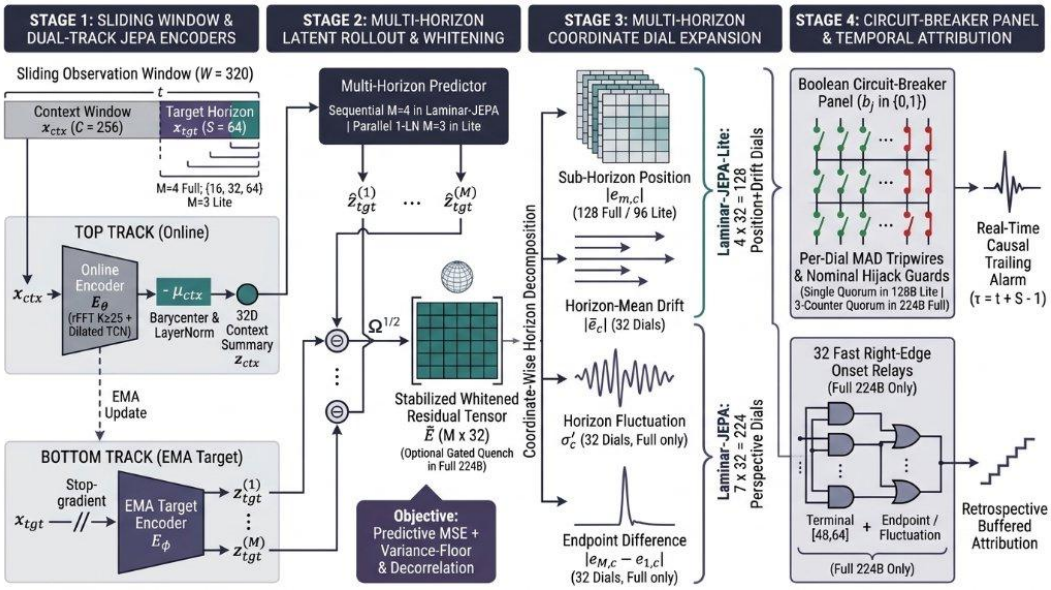


Fig. 1. Overview of the four-stage *Laminar-JEPA* and compact *Laminar-JEPA-Lite* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K \geq 25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout (M=4 sequential in full *Laminar-JEPA*; M=3 parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal (τ=t+S−1) or prefix-buffered alarms.

a two-layer GELU head and blended with the deep TCN output through a bounded gate ($g \leq 0.5$).

2) *Barycenter centering (removes common offsets):* During training, a running mean of the context summaries $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (update rate 0.05, shared with E_ϕ) is subtracted from both context and target summaries before the final LayerNorm. Centering removes offsets learned in training; it does not guarantee invariance to unseen drift.

B. Stage 2: Multi-Horizon Prediction

The simplest JEPA detector, the single-horizon baseline *TS-JEPA*, uses a two-layer multi-layer perceptron (MLP) to predict one summary $\hat{z}_{\text{tgt}}$ of the whole 64-step target window, trained with predictive mean-squared error (MSE) and a VICReg-style log-determinant penalty ($\beta = 0.05$) [21]. Summarizing all 64 steps in one vector can average away a short fault. *Laminar-JEPA* therefore predicts how the target unfolds: it summarizes $M = 4$ nested prefixes of the target window, the first $S_m \in \{16, 32, 48, 64\}$ steps. Starting from z_{ctx} plus learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, a recurrent predictor P_{sub} produces $\hat{z}_{\text{tgt}}^{(1)}, \ldots, \hat{z}_{\text{tgt}}^{(M-1)}$ in sequence, and an anchor head P_θ predicts the full-window summary $\hat{z}_{\text{tgt}}^{(M)}$. The loss gives the full window weight $w_M = 1/2$ and each shorter prefix $w_m = 1/(2(M-1))$ (1/6 in the full model, 1/4 in Lite), and adds a penalty $\boldsymbol{\Delta}_\gamma$ that keeps the summaries from collapsing to a constant:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D} \left(\left\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\left(z_{\text{tgt}}^{(m)}\right) \right\|_2^2 + \left\| \boldsymbol{\Delta}_\gamma\left(\hat{z}_{\text{tgt}}^{(m)}\right) \right\|_F^2 \right), \tag{1}$$

where $\text{sg}(\cdot)$ stops gradients through E_ϕ and the penalty acts on the predicted summaries of each batch. Log-determinant penalties need batches larger than D; $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ works on small batches. Its diagonal entries $\sqrt{\max(0, \gamma - \sigma_d)}$ push each coordinate's standard deviation above $\gamma = 0.75$ (σ_d from biased batch variance plus 10^{-4}) and its off-diagonals $\Sigma_{dj}/\sqrt{2}$ penalize sample covariance (divisor $B-1$, both symmetric entries counted), so $\frac{1}{D}\|\boldsymbol{\Delta}_\gamma\|_F^2 = \frac{1}{D}\sum_{d=1}^{D} \max(0, \gamma - \sigma_d) + \frac{1}{2D}\sum_{d\neq j} \Sigma_{dj}^2$. At $B = 16 < D = 32$, this lifts target effective rank $\text{RankMe}(z_{\text{tgt}})$ [49] from 11.6 to 19.8.

C. Stage 3: Residual Whitening and Dial Expansion

Stage 3 turns prediction errors into dials. The signed errors $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ for the $M = 4$ horizons form a 4×32 error matrix $\mathbf{E}(t)$. Its entries have different natural sizes and are correlated, so they are first passed through a *damped whitening* step fitted on healthy training windows, which decorrelates them and brings well-estimated directions close to unit variance. We flatten $\mathbf{E}(t)$ into a vector of length MD (128 in the full model, 96 in Lite), estimate its mean and unbiased covariance on the $\mathbf{X}_{\text{fit}}$ windows, and with $\hat{\boldsymbol{\Sigma}}_{\text{res}} = \mathbf{U}\,\text{diag}(\lambda_k)\mathbf{U}^\top$ ($\mathbf{U}, \boldsymbol{\Omega} \in \mathbb{R}^{MD\times MD}$) use the damped precision matrix $\boldsymbol{\Omega} = \mathbf{U}\,\text{diag}\big(\lambda_k/(\lambda_k + 10^{-3})^2\big)\mathbf{U}^\top$. Multiplying centered errors by $\boldsymbol{\Omega}^{1/2}$ gives whitened errors $e_{m,c}^{\text{white}}(t)$ whose training covariance has eigenvalues $\lambda_k^2/(\lambda_k + 10^{-3})^2$: close to 1 for well-estimated directions but deliberately below 1 for directions with almost no training variance, which are attenuated rather than amplified. The whitening is therefore approximate, and the damping constant 10^{-3} is fixed across all cohorts.

1) *Stabilization on a separate healthy block $\mathbf{X}_q$:* A coordinate whose error grows between training and later healthy data would be inflated by whitening. We detect such coordinates on a separate healthy block $\mathbf{X}_q$, never on the block $\mathbf{X}_{\text{cal}}$ that later sets alarm thresholds (Section V-A), and blend them toward a simpler per-coordinate scaling, the unwhitened error divided by its root-mean-square on $\mathbf{X}_q$, $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/\text{RMS}_q(e_{m,c})$. The blend starts when the whitened

RMS on $\mathbf{X}_q$ exceeds 1.10, and all weights are frozen before $\mathbf{X}_{\text{cal}}$ is used:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e^{\text{white}}_{m,c}(t) + (1 - w_{m,c})\, \hat{e}^{\text{raw}}_{m,c}(t), \qquad w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}^{\text{white}}_q(m,c) - 1.10\big)^2}. \tag{2}$$

The full *Laminar-JEPA* can also run an optional gradient-free *weight-space quench*, which edits the few predictor outputs that drift most on healthy data and keeps the edit only if false alarms fall. It uses only $\mathbf{X}_{\text{fit}}$ and $\mathbf{X}_q$ (legacy fixed-epoch development runs without $\mathbf{X}_q$ use $\mathbf{X}_{\text{cal}}$; Table I). Coordinates are ranked by a median-absolute-deviation (MAD) standardized instability score that combines mean and scale shifts from $\mathbf{X}_{\text{fit}}$ to $\mathbf{X}_q$, drift between the halves of $\mathbf{X}_q$, and residual autocorrelation, with a negative term for isolated spikes. For the top $k \in \{4, 2\}$ coordinates, four steps run in order. *(Q1)* Damp the selected predictor output rows and LayerNorm gains and shift $\mathbf{p}_m$ toward the $\mathbf{X}_q$ means; through the shared 32D LayerNorm this also moves unselected coordinates (Section VI-A). *(Q2)* Move residual centers 75% of the way toward the post-edit $\mathbf{X}_q$ means. *(Q3)* Scale off-diagonal precision couplings by 0.15 and rescale whitened $\mathbf{X}_q$ RMS. *(Q4)* Keep the edit (preferring $k = 2$ on ties) only if the $\mathbf{X}_q$ half-split FPR improves while at least 92% of the pre-edit quorum thresholds are preserved; then freeze all parameters.

2) *Expansion into $7D = 224$ dials:* For each of the 32 coordinates, the four horizon errors describe how the prediction error evolves across the target window. Borrowing the mean-plus-fluctuation split of Reynolds averaging [35] purely as algebra, we split them into a horizon mean $\bar{e}_c(t) = M^{-1}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and a zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$. Four views of each coordinate give $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) $4D = 128$ *sub-horizon positions* $|\tilde{e}_{m,c}(t)|$, the error size at each horizon; (b) $D = 32$ *horizon-mean drifts* $|\bar{e}_c(t)|$, a persistent offset shared by all horizons; (c) $D = 32$ *horizon fluctuations* $d_{\text{fluc},c}(t) = \sqrt{M^{-1}\sum_{m=1}^{M} F^2_{m,c}(t)}$, how much the error changes across horizons; and (d) $D = 32$ *endpoint differences* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, error that appears late in the window, as at a fault onset.

The proposition below states two exact properties of this split. Part (i) says that drift and fluctuation measure non-overlapping parts of the position energy, and that the endpoint dial cannot exceed a multiple of the fluctuation dial. Part (ii) says the fluctuation pattern lives in at most $M - 1$ directions and defines R_{LT}, which is 1 when one pattern dominates and $1/(M-1)$ when the energy is spread evenly. These are geometric facts about the representation, not accuracy guarantees.

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *For $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) with horizon mean $\bar{\mathbf{e}}(t)$ and fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M \bar{\mathbf{e}}(t)^\top \neq \mathbf{0}$: (i) for every coordinate c, $M^{-1}\sum_{m=1}^{M} |\tilde{e}_{m,c}(t)|^2 = |\bar{e}_c(t)|^2 + d^2_{\text{fluc},c}(t)$ and $d_{\text{accel},c}(t) \leq \sqrt{2M}\, d_{\text{fluc},c}(t)$; and (ii) the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^\top \in \mathcal{S}^M_+$ has ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ whose scale-invariant dominant-mode share for any $p > 0$ [50] satisfies*

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda^p_M}{\sum_{m=1}^{M} \lambda^p_m} \in \Big[\frac{1}{M-1}, 1\Big], \tag{3}$$

reaching $\frac{1}{M-1}$ iff energy is isotropic across all $M-1$ active modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$) and 1 iff $\mathbf{F}(t)$ has rank one. Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).

Proof. Zero column sums ($\sum_{m=1}^{M} F_{m,c} = 0$) cancel the cross term in $M^{-1}\sum_{m=1}^{M}(\bar{e}_c + F_{m,c})^2$, and $(F_{M,c} - F_{1,c})^2 \leq 2(F^2_{M,c} + F^2_{1,c}) \leq 2M\, d^2_{\text{fluc},c}$ gives (i). For (ii), $\mathbf{F}^\top \mathbf{1}_M = \mathbf{0}_D$ implies $\mathbf{G}\mathbf{1}_M = \mathbf{0}_M$ ($\lambda_1 = 0$), while $\|\mathbf{F}\|^2_F > 0$ implies $\lambda_M > 0$. Summing $0 \leq \lambda^p_m \leq \lambda^p_M$ over $m \in \{2, \ldots, M\}$ yields $\lambda^p_M \leq \sum_{m=2}^{M} \lambda^p_m \leq (M-1)\lambda^p_M$; inverting and multiplying by λ^p_M gives (3) and its equality conditions, and nonzero scalar rescaling cancels. □

D. Stage 4: Circuit-Breaker Panel

Instead of adding all 224 dials into one score, Stage 4 (Algorithm 1) gives each dial its own threshold, learned from the healthy calibration block. A dial *trips* when it exceeds its threshold, and an alarm fires when enough dials trip at once. Because all dials come from the same 4×32 error matrix, their trips are correlated (median effective rank 65 of 224 on the development cohort; the variance of the trip count is $7.2\times$ that of independent dials). We therefore do not assume independence; the alarm quorums are set from the trip counts actually observed on healthy calibration data:

1) *Per-dial thresholds and hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized by its calibration median m_j and scaled MAD $s_j = 1.4826\ \text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ (the 10^{-4} floor handles zero-MAD dials), giving $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. Its tripwire τ_j is the 98.5th calibration percentile of z_j, widened by $\gamma^{\text{cal}}_j \geq 1$ when its calibration tail grows (second-half $q_{0.98}$ above $1.35\times$ the first half) and by $\gamma^{\text{sc}}_j \geq 1$ when its scale exceeds $1.875\times$ its perspective-group median:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma^{\text{cal}}_j\, \gamma^{\text{sc}}_j. \tag{4}$$
2) *32 onset relays:* When a sudden fault first enters the right edge of the target window, the short prefixes ($m \in \{1, 2\}$) still cover healthy data, so few position dials trip. To catch this moment, each coordinate c has stricter 99.3rd-percentile edge tripwires ($\gamma^{1.25}$ guard) and fires a relay $r_c(t) \in \{0, 1\}$ when its full-window position dial ($m = 4$) trips together with its endpoint, fluctuation, or $m = 3$ dial.
3) *Alarm timing and quorum voting:* Under *causal attribution*, the trips from window $[t, t+S)$ are reported at its right end $\tau = t + S - 1$, the earliest moment all of its data exist; this fixes a reporting lag of $S - 1$ steps, separate from the detection delay $\Delta t = t_{\text{alm}} - t^*$ after fault onset t^*. Under *buffered attribution*, used for after-the-fact review, each trip is assigned to the steps it describes: position trips to their prefix $[t, t+S_m)$, drift and fluctuation trips to $[t, t+S)$, and endpoint trips to $[t+S_3, t+S)$. Counting active trips at each step gives $N_{224}(\tau)$, the position sub-count $N_{128}(\tau) \subset N_{224}(\tau)$, and the relay count $N_{\text{edge}}(\tau)$. For ranking metrics, the continuous score

$N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ counts position trips and relays twice (fixed on development streams before held-out testing; Section VI-A). The binary alarm fires when any count exceeds its quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \ \vee \ N_{128}(\tau) > K_{128} \ \vee \ N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \tag{5}$$

with quorums set on calibration counts: $K_{224} = \max(1.12\, q_{0.995}, 5.0)$, $K_{128} = \max(1.12\, q_{0.995}, 3.5)$, and $K_{\text{edge}} = \max(1.25\, q_{0.999}, 2.5)$. All quantiles use linear interpolation, the safety factor is applied before the floor, quorums stay real-valued, and both trips and alarms use strict inequalities.

E. *Compact Variant:* Laminar-JEPA-Lite

Laminar-JEPA-Lite keeps the same four stages but makes three simplifications, each supported by the ablations in Section VI. Together they cut trainable parameters by 35%–52% (141.9k at $K < 25$ and 222.3k at $K \geq 25$, vs. 295.8k), speed up windowed GPU inference by 2.43× (3.14× at $K < 25$), and need no weight editing:

1) *Smaller encoder:* A fifth dilated block blurred short-lived anomalies without making the summaries richer, so *Laminar-JEPA-Lite* uses four TCN blocks (dilations $\{1, 2, 4, 8\}$) and 64D pooling heads. The frequency-band front-end is used only on high-channel streams ($K \geq 25$), where it helped; smaller streams use a direct causal projection.
2) *Parallel three-horizon predictor:* One residual two-layer GELU MLP predicts $M = 3$ prefixes ($S_m \in \{16, 32, 64\}$) in parallel from $z_{\text{ctx}} + \mathbf{p}_m$, with a single final LayerNorm. This removes the lag of sequential prediction and the coupling between stacked normalizations, and shrinks $\mathbf{\Omega}$ to 96×96.
3) *128-dial panel with one quorum:* Only the 96 position dials $|\tilde{e}_{m,c}(t)|$ and the 32 drift dials $|\bar{e}_c(t)|$ are kept, each calibrated by (4). Fluctuation and endpoint dials, onset relays, and quenching are removed. Steps are ranked by the plain count $N_{128}(\tau)$, and the alarm uses one quorum, $K_{\text{Lite}} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 3.5\big)$.

1D single-score controls: To measure what the breaker panel adds over a single score on the same backbone (Section VI), the 1D controls pool a residual vector into one number: an equal blend of its ℓ_2 norm and a softmax-weighted coordinate RMS ($\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA pools $\mathbf{e}(t) \in \mathbb{R}^{32}$; 1D Multi-Horizon JEPA combines the pooled full-tensor and peak-horizon scores by root-mean-square (with optional R_{LT} modulation $\alpha_{\text{LT}} \in \{0.12, 0\}$). Both are thresholded with SPOT [8], [38], [51].

V. Experimental Protocol and Evaluation Metrics

A. Setup and Calibration Protocol

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in$ {Full, Lite}, attribution mode $\in$ {causal, buffered}.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$.

Phase 1: Disjoint Training, Whitening & Adaptation ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q$)
Split $\mathbf{X}_{\text{nom}}$ chronologically into S-isolated ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q, \mathbf{X}_{\text{cal}}, \mathbf{X}_{\text{hold}}$) and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
  Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
  Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (1) and $\phi \leftarrow m\phi + (1-m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit $(\bar{\boldsymbol{\mu}}_{\text{res}}, \mathbf{\Omega} \in \mathcal{S}_+^{MD})$ on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
  Rank drifting coordinates on $(\mathbf{X}_{\text{fit}}, \mathbf{X}_q)$; run Steps Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff $\mathbf{X}_q$ split and 0.92-quorum gates pass.
end if
Fit and freeze stabilization scales and blending weights $w_{m,c}$ (2) on $\mathbf{X}_q$.

Phase 2: Untouched Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Compute frozen stabilized residuals $\tilde{\mathbf{E}}_{\text{cal}}$ on untouched $\mathbf{X}_{\text{cal}}$ via (2).
if tier = Full **then**
  Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 right-edge onset relays.
  Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5).
else ▷ Preferred Deployment Tier: Laminar-JEPA-Lite
  Extract $4D = 128$ position-plus-drift dials $(|\tilde{e}_{m,c}|, |\bar{e}_c|)$, fit τ_j (4), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 3.5)$.
end if

Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t - C,\ t + S)$ **do**
  Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
  Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
  Emit $\hat{y}(\tau) \in \{0, 1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (5) or K_{Lite}.
end for

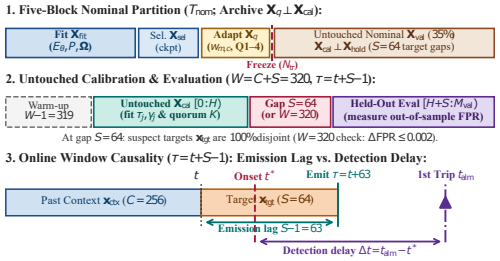


Fig. 2. Disjoint nominal evaluation protocol (C=256, S=64, W=320): (Row 1) five-block archive partition into S-isolated weight-fitting ($\mathbf{X}_{\text{fit}}$), checkpoint-selection ($\mathbf{X}_{\text{sel}}$), and nominal adaptation ($\mathbf{X}_q$) prefixes versus $\mathbf{X}_{\text{val}}$, (Row 2) untouched threshold calibration ($\mathbf{X}_{\text{cal}}$) and held-out nominal evaluation ($\mathbf{X}_{\text{hold}}$), and (Row 3) structural emission lag ($S-1$) versus empirical detection delay (Δt).

a) Disjoint Calibration Protocol: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Table I) into a model-fitting prefix $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = T_{\text{nom}} - \lfloor 0.35\, T_{\text{nom}} \rfloor$) and a held-out nominal suffix $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$: (1) *Stage A (Disjoint weight fitting, checkpoint selection, and adaptation):* Reusing cali-

TABLE I
NOMINAL PARTITIONING AND CALIBRATION PROTOCOL BY COHORT

Cohort (Tables)	Weights (%)	Ckpt. Sel.	Adapt. / Quench	Thresholds
Dev. 19 Cache (IIa,b, III)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	None (unblended)	$\mathbf{X}_{\text{cal}}$
Dev. 33 Suite (IIc, IV, V)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	Shared $\mathbf{X}_{\text{cal}}$	$\mathbf{X}_{\text{cal}}$
Dev. 19 4-Block (Sec. V-B)	$\mathbf{X}_{\text{fit}}$ (50)	Fixed ep.	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Archive 118/141 (VI)	$\mathbf{X}_{\text{fit}}$ (43)	$\mathbf{X}_{\text{sel}}$	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
130 Screen (Sec. VII-A)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	None	$\mathbf{X}_{\text{cal}}$
Held-Out / Test ($\mathbf{X}_{\text{hold}}$, $\mathbf{X}_{\text{test}}$)	Frozen	Frozen	Frozen	Frozen

Roles: development rows are design-stage evidence (causal and buffered); the archive is the primary evidence (causal and buffered); the 130-stream single-seed screen (buffered values reported) is context for external baselines only and is not paired with the archive.

bration windows for early stopping, blending (2), or quenching can make nominal thresholds optimistic. Archive benchmarks (Table VI) therefore use a disjoint *five-block protocol* ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$) inside $\mathbf{X}_{\text{tr}}$: $\mathbf{X}_{\text{fit}}$ ($\approx 43\%$ of T_{nom}) fits standardization, network weights, $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$; $\mathbf{X}_{\text{sel}}$ ($N_{\text{sel}} \geq 96$ windows) selects the checkpoint; and $\mathbf{X}_q$ ($N_q \in [160, 2{,}048]$ stride-1 windows isolated from $\mathbf{X}_{\text{sel}}$ and $\mathbf{X}_{\text{val}}$ by $S = 64$ steps) fits and freezes (2) and optional quenching (Steps Q1–Q4) before $\mathbf{X}_{\text{cal}}$ is touched. Fixed-epoch development runs (Table I) omit $\mathbf{X}_{\text{sel}}$, and the 19-stream cache skips (2) and quenching. A four-block check ($\mathbf{X}_{\text{fit}} \to \mathbf{X}_q \to \mathbf{X}_{\text{cal}} \to \mathbf{X}_{\text{hold}}$) on the 19 development streams gives causal FPR_{hold} of 0.96% (*Lite*) and 1.05% (*Laminar-JEPA*), matching Table IV, though the smaller 50% fitting block lowers causal F1 ($0.133 \to 0.071$, $0.144 \to 0.076$). On short histories, *two-fold cross-fitting* swaps the halves of $\mathbf{X}_{\text{val}}$ and requires $\hat{y}^{(1)} \wedge \hat{y}^{(2)}$ (Section VII-E). (2) *Stage B (Untouched calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ in $\mathbf{X}_{\text{val}}$ give $M_{\text{val}} = N_{\text{val}} - W + 1$ indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines $\mathbf{X}_{\text{cal}} = [0 : H)$ and $\mathbf{X}_{\text{hold}} = [H + S : M_{\text{val}})$, separated by an $S = 64$-step gap $[H : H + S)$ so target observations never overlap (a $G = 320$ gap shifts causal FPR by < 0.25 points). On $\mathbf{X}_{\text{cal}}$ alone, the frozen model fits all dial scales, guards, tripwires (4), and quorums (5) (or τ_{SPOT} for 1D baselines), frozen on $\mathbf{X}_{\text{hold}}$ and $\mathbf{X}_{\text{test}}$. We separate raw count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ via Sokal estimators (τ_{int} up to 206; Section VII-E).

b) Statistical Inference: We report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference: Dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For the four focal causal contrasts on the 118 untouched streams (*Laminar-JEPA-Lite* vs. Multi-Horizon JEPA (1D) and TS-JEPA, on Point-F1 and ER), Benjamini–Hochberg (BH) [41] adjusted p-values are 0.014 and 3.6×10^{-4} (stream-level Point-F1) and 0.0078 and 0.012 (dataset sign-flip ER, positive on $9/9$ and $8/9$ datasets); other quoted p-values are raw and descriptive, and non-Exathlon subsets ($n = 101$, 8 datasets) verify sensitivity to archive composition. (2) *12-fold LODO audit:* Across all 12 LODO folds of the 141-stream archive, *Laminar-JEPA-Lite* keeps a positive gain on causal ER ($+0.099$ to $+0.141$ vs. Multi-Horizon 1D), buffered Point-F1, and causal Point-F1 vs. TS-JEPA ($\max p = 0.0054$); its causal Point-F1 gain over Multi-Horizon 1D holds in $9/12$ folds.

B. Cohorts and Baselines

Laminar-JEPA (w/o Quench) omits Steps Q1–Q4. Causal and buffered modes share identical splits per cohort (except a single-horizon 130-stream buffered screen in Section VII-A): (1) *Development cohorts (design-stage, Table I):* Because these streams were inspected during design and the 33-stream suite shares $\mathbf{X}_{\text{cal}}$ for stabilization and thresholds, they compare design choices rather than out-of-sample calibration. The 19-stream cache (6 datasets) tests the dial ladder, panel components, and conformal/SPC banks on shared unquenched residuals (Table II(a,b), Table III); the 33-stream benchmark (7 datasets: 19 non-GHL in Table II(c) plus 14 GHL in Table IV) compares all baselines. Tripwire and quench constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary evidence):* Removing those 23 tuning streams from the 141-stream archive (12 datasets, five-block protocol) leaves 118 untouched streams across 9 datasets (101 non-Exathlon across 8 datasets; Table VI), plus a single-seed Q1–Q4 isolation audit (Section VI-A).

Three baseline tiers: (i) *Matched-backbone controls* (TS-JEPA, Multi-Horizon JEPA (1D)) fix the representation and test one SPOT score against the panel. (ii) *System-level aggregators* fix the whitened residuals and replace the panel with maximum z, top-k voting, ℓ_2^2 conformal, Fisher/Stouffer p-pooling, multiplicity-controlled conformal banks, or SPC charts, calibrated to the panel's $\mathbf{X}_{\text{cal}}$ alarm rate unless noted (Tables III, VI). (iii) *External baselines* (TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], DCdetector [17]) share splits, training-only scaling, gaps, and SPOT ($u = p_{98}$, $q = 10^{-3}$), using 206k–649k parameters (panels add none). On the archive, every JEPA backbone receives the same budget: seeds $\{42, 123, 456\}$ reset all random number generators before each backbone, and each seed trains independently with AdamW (10^{-3}, weight decay 10^{-4}, cosine schedule) for up to 12 epochs with its own early stopping on $\mathbf{X}_{\text{sel}}$ (patience 4). τ_{int} is estimated separately on each calibrated statistic's own $\mathbf{X}_{\text{cal}}$ series with Sokal's automatic window ($c = 5$), and a q-quantile has $N_{\text{eff}}(1 - q)$ expected effective exceedances. Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (0.15 ms/window *Lite*, 0.38 ms *Laminar-JEPA*; 188 MB peak); code: https://github.com/quaid281/NCAD.

C. Metrics

Every alarm $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$ is scored step by step against $y(\tau) \in \{0, 1\}$ without point adjustment [11]. *Causal* numbers show what an operator sees live at step τ; *buffered* numbers reflect retrospective review after the target window finishes and are not live performance. Headline values average seeds per stream before averaging across streams (or unweighted across datasets in dataset-macro Buffered Point-F1, DS-F1):

1) *False-alarm rates (FPR_{hold}, FPR_{test}):* Fraction of healthy steps flagged on the untouched pre-test check block $\mathbf{X}_{\text{hold}}$ ($[H + S : M_{\text{val}})$) and on healthy test steps ($y(\tau) = 0$).

Because a few drifting streams skew the mean, we also report stream medians and 90th percentiles (p90).

2) *Step-level accuracy (Point-F1, precision-recall (PR) AUC):* Point-F1 (Pt-F1) is the harmonic mean of step precision (fraction of alarm steps that are truly faulty) and step recall (fraction of faulty steps flagged). Without point adjustment, Pt-F1 is naturally modest (0.05–0.25, not 0.90): residuals peak near regime changes rather than across every step of a long fault, and when faults occupy a small fraction π of the stream, even a 1% false-alarm rate adds many false-positive steps (alarming at every step scores $\mathrm{F1}_{\mathrm{all}} = 2\pi/(1+\pi)$). PR-AUC (Buf. PR when buffered) scores ranking quality across all thresholds without inflation from healthy steps.
3) *Episode catch rate and speed (ER, Delay Δt):* Event recall (ER) is the fraction of contiguous faulty episodes ($y = 1$ runs) with at least one alarm; it must be read with FPR because random alarms at rate p still hit long episodes at rate $\mathrm{ER}_{\mathrm{rand}}(p)$. Detection delay $\Delta t = t_{\mathrm{alm}} - t^*$ counts steps from fault onset t^* to the first in-segment alarm over *detected* episodes (distinct from the $S-1$ causal lag), so catching an extra late-developing fault can raise mean Δt.
4) *Budget controls and score diagnostics: Auto.* uses the deployed healthy-only threshold (quorums or SPOT), $q^* = 0.05$ sets thresholds at the 95th $\mathbf{X}_{\mathrm{cal}}$ percentile (q_{1-q^*}), and diagnostic controls $f^* \in \{0.01, 0.05\}$ use test labels post-hoc to match a 1% or 5% $\mathrm{FPR}_{\mathrm{test}}$. On nominal scores, excess kurtosis γ_2 ($\gamma_2 = 0$ for Gaussian tails) measures $\mathbf{X}_{\mathrm{cal}}$ spike heaviness, and the two-sample Kolmogorov–Smirnov (KS) distance $D_{\mathrm{KS}} \in [0,1]$ ($\mathbf{X}_{\mathrm{cal}}$ vs. $\mathbf{X}_{\mathrm{hold}}$) measures healthy drift.

VI. Ablation Studies and Analysis

A. Ablation 1: Panel Components and Quenching

Table II(a),(b) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Scientific role of the 224-dial reference vs. compact panels:* More dials do not make a better operational detector; we keep 224-dial *Laminar-JEPA* as a diagnostic reference because scoring all four views on one residual tensor isolates which views aid live causal versus buffered attribution. Against 1D scoring on this cache, 224 dials raise causal Point-F1 from 0.013 to 0.071 ($p = 0.0023$; buffered $+0.033$, $p = 0.53$), whereas pruning fluctuation, endpoint, and relay dials to 160 position-plus-drift dials raises causal Point-F1 to 0.098, ER to 59.1%, and cuts delay ($90.1 \to 74.8$ steps) and $\mathrm{FPR}_{\mathrm{hold}}$ ($0.76\% \to 0.55\%$) while lowering buffered Point-F1 ($0.224 \to 0.198$). Retraining on the 118 untouched archive streams (Table VI(b)) tests 224-dial *Laminar-JEPA*, same-run 160-dial *Laminar-JEPA*, and 3-horizon *Laminar-JEPA-Lite* (128 dials) on identical splits: pruning $224 \to 160$ dials cuts buffered $\mathrm{FPR}_{\mathrm{hold}}$ by 6.2 points ($19.2\% \to 13.0\%$, $p = 3.7 \times 10^{-15}$) at near-identical causal Point-F1 (0.073 vs. 0.074), while *Laminar-JEPA-Lite* matches or exceeds both (0.078 causal, 0.154 buffered Point-F1) at lower cost.
2) *Relays, PR-AUC score weighting, quorum rules, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 ($0.071 \to 0.068$), and replacing the three-counter quorum (5) with one 224-dial quorum lowers causal Point-F1 to 0.063 ($\mathrm{FPR}_{\mathrm{hold}} = 0.0055$; Table III). Replacing the double-position continuous score $N_{\mathrm{tot}} = N_{224} + N_{128} + 2N_{\mathrm{edge}}$ with unweighted $N_{224}+N_{\mathrm{edge}}$ or N_{224} leaves binary alarms unchanged and changes PR-AUC by at most 0.004 on this cache and 0.0016 on the 118 untouched streams (*Laminar-JEPA-Lite* uses unweighted N_{128}). Removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 (70.1-step delay) on this cache when nominal drift is mild, so the guards are a safeguard against drift rather than a source of accuracy. No panel component removes the $S-1$ reporting lag.
3) *Quenching step isolation and LayerNorm coupling:* We tested each quench step in isolation on single-seed *Laminar-JEPA* checkpoints across all 118 untouched streams (unquenched: causal $\mathrm{FPR}_{\mathrm{hold}} = 6.76\%$, F1 $= 0.0901$). Damping $k = 4$ predictor rows (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS through 32D LayerNorm coupling and does not lower $\mathrm{FPR}_{\mathrm{hold}}$ (7.06%). Recentering and decoupling (Steps Q2–Q3) lower it to 6.71% but cut causal F1 to 0.0826. The split gate (Step Q4, accepted on $39/118$ streams) reaches 6.62% with F1 unchanged (0.0904). Because ungated edits hurt $k = 2$ causal F1 on $11/19$ development streams ($0.141 \to 0.049$) and $k \in \{8, 12\}$ cuts PR-AUC ($0.292 \to 0.223$), *Laminar-JEPA-Lite* omits weight editing altogether.

Marginal, multiplicity-controlled, and system-matched detector banks: Simple per-coordinate alarms either fire constantly without a quorum, almost never fire under multiplicity correction on short calibration sets, or lose their calibration out of sample. We compare three calibration regimes on the same 19-stream whitened residuals without test-label tuning (Table III): (1) *Per-coordinate marginal union:* Thresholding 128 coordinates individually at $q_{0.985}$ or even at the panel's 0.34% rate without a quorum compounds false alarms across coordinates ($\mathrm{FPR}_{\mathrm{hold}} = 66.5\%$ and 40.6%). (2) *Family-wise and false discovery rate (FDR) multiplicity control:* Applying Šidák (0.5%) across 128 coordinates or BH FDR ($q = 0.005$) across 224 dials hits the discrete conformal p-value floor $(N_{\mathrm{cal}} + 1)^{-1} > 3.9 \times 10^{-5}$, paralyzing causal recall (F1 ≤ 0.001). (3) *System-level matched nominal alarm rate:* Calibrating the aggregated statistic directly to each stream's nominal panel rate (0.34%) shows that discrete conformal $\min_j p_j$ still jumps to $\mathrm{FPR}_{\mathrm{hold}} = 21.1\%$ out of sample from boundary ties, and recalibrated Mahalanobis conformal, Hotelling T^2, MEWMA, and CUSUM inflate to 27.9%–58.6%. Among stable rules on this cache, the 160-dial panel (F1 $= 0.098$, $\mathrm{FPR}_{\mathrm{hold}} = 0.55\%$) beats 128-coordinate top-k (0.068), $\max_j z_j$ (0.056), and ℓ_2^2 conformal (0.072), whereas Fisher pooling reaches higher Point-F1 (0.109; Stouffer 0.095) at nearly double the held-out FPR (1.05% vs. 0.55%). Archive comparisons of these aggregators on frozen *Laminar-JEPA-Lite* residuals appear in Section VII-E.

TABLE II
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder ($1 \rightarrow 224$ Dials, n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Perspective Expansion Stage	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Component / Quorum Variant	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (2). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	Auto.	q^*=0.05	f^*=0.01	f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

Panel (c) (5 seeds): γ_2 = excess kurtosis on $\mathbf{X}_{\text{cal}}$; D_{KS} = KS drift ($\mathbf{X}_{\text{cal}}$ vs. $\mathbf{X}_{\text{hold}}$); Auto. = deployed healthy-calibrated threshold; q^*=0.05 = 95th-percentile $\mathbf{X}_{\text{cal}}$ threshold; f^* = post-hoc diagnostic FPR_{test} control.

TABLE III
COORDINATE DETECTOR-BANK REGIMES ON CACHED RESIDUALS

Detector & Calibration Regime	Pt-F1 ↑	ER ↑	FPR_{hold} ↓	FPR_{test} ↓
Multi-horizon circuit-breaker panels (nominal quorum)				
160-dial pos.+drift (single quorum)	**0.098**	0.591	0.0055	0.0087
224-dial panel (deployed)	0.071	0.564	0.0076	0.0150
224-dial single quorum (no relay)	0.063	0.564	0.0055	0.0088
1. Per-coordinate marginal union (no quorum, 128 coords)				
Per-coord. $q_{.985}$ union	0.291	0.913	0.6651	0.7353
Per-coord. .0034 marginal union	0.303	0.874	0.4057	0.4765
2. Family-wise / FDR multiplicity control (α, q=.005)				
Per-coord. conformal (Šidák)	0.000	0.000	**0.0000**	**0.0000**
224-dial conformal BH FDR	0.000	0.026	**0.0000**	**0.0000**
3. System-level matched nominal rate (mean .0034 on $\mathbf{X}_{cal}$)				
Raw 128-coord. top-k vote	0.068	0.564	0.0052	0.0096
Continuous 128-coord. $\max_j z_j$	0.056	0.658	0.0078	0.0147
Discrete conformal $\min_j p_j$	0.265	0.843	0.2111	0.2831
Multivar. conf. ($\|\mathbf{e}_{\text{white}}\|_2^2$)	0.072	0.608	0.0062	0.0095
Fisher p-pool (224 dials)	0.109	0.643	0.0105	0.0183
Maximum 224-dial z	0.083	0.652	0.0093	0.0157
Hotelling T^2 (recalibrated)	0.283	0.816	0.2900	0.4831

19 streams × 3 seeds, causal means. System-matched rules match each stream's nominal panel alert rate after aggregation (Stouffer .095 F1, .0087 FPR_{hold}; 224-dial top-k .070, .0077; Mahalanobis conf. .2789, CUSUM .3457, MEWMA .5862 FPR_{hold}).

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

A frozen or flatlined sensor barely moves the latent summary: terminal LayerNorm spreads the small change across all 32 coordinates, so on `SMAP P-3` a flatline trips 12.5 dials versus 62.1 for a variance-matched harmonic flip, and a single faulty sensor is further diluted by the $K-1$ healthy channels. Extending an initial six-stream study, we evaluate a causal channel branch across all 19 multivariate development streams ($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\text{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows (floor 6.0 MAD, 1.25× half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel we inject three nonoverlapping 200-step faults per stream into clean test segments (57 events per class): variance collapse to the training mean, an alternating $\pm 1.35\sigma$ prediction shock, sensor freezing, saturation at $\max(q_{.995}, \mu + 2.2\sigma)$, a linear 2.5σ degradation, and a 1.5σ bias. The variance-collapse union lifts injected-event recall for variance collapse, freezing, and saturation from 63.2%–82.5% to 93.0%–94.7%, and the full channel branch lifts recall on shocks, gradual degradation, and bias from 59.6%–75.4% to 80.7%–96.5%. However, on the *unmodified* test streams, causal FPR_{test} rises from 1.5% (standalone panel, F1 = 0.071) to 6.8% (variance-collapse union, F1 = 0.070) and 19.4% (full branch, F1 = 0.129). Synthetic single-channel injections do not establish detection on natural sensor failures, and the full union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) shows a channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 0.085 → 0.410; K=25 `SMAP P-3` causal F1 0.212 → 0.321), but lowers F1 on both K=9 streams (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Across the 16 matched-budget arms of Table II(c) (development streams; contributing factors, not universal causes), raw-observation forecasting and autoencoding share heavy tails

($\gamma_2 \approx 6.2$), large drift ($D_{\text{KS}} \approx 0.45$), and $\approx 25\%$ causal FPR; a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR but keeps drift high ($D_{\text{KS}} = 0.376$); learned latent targets and whitening halve drift ($D_{\text{KS}} = 0.179$–0.214, $p \leq 0.001$); and PCA and Isolation Forest keep low causal FPR but reach only 0.017 and 0.041 causal Point-F1. The panels' gains do not come from lighter tails alone: *Laminar-JEPA-Lite* keeps roughly the same kurtosis as Multi-Horizon JEPA (1D) ($\gamma_2 = 2.33$ vs. 2.30).

VII. Results and Discussion

A. Design-Stage Benchmark and Scope Boundaries

Following Section V-B, Table IV reports exploratory design-stage results on the 33-stream development suite (where stabilization and thresholds share $\mathbf{X}_{\text{cal}}$). Primary evidence is the five-block archive (Section VII-E). A separate single-seed 130-stream buffered screen gives context only and is not paired with the archive: TS-JEPA reaches Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082.

Configuration: On the 33-stream suite both tiers reach similar buffered Point-F1 (0.244 vs. 0.242), but *Laminar-JEPA-Lite* halves buffered FPR_{hold} (6.1% vs. 12.8%) and leads macro Point-F1 (0.171 vs. 0.133).

Slow-ramp boundary: A $C = 256$-step encoder cannot see drift slower than its window. Context and target summaries move together, yielding 0.000 Point-F1 on all 14 GHL slow-ramp streams. Injecting linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and length $L \in \{64, 256, 1024, 4096\}$ (15 events per cell) maps the cutoff: steep multi-channel ramps ($L = 64$, $a \geq 1\sigma$) reach 86.7%–88.7% ER at 46–54-step median delay (5.7–7.7 points above background chance), $L = C$ falls to 0.5–2.5 points above chance, and for $L \geq 1024$ ($4C$) chance-corrected detection vanishes ($[-4.8, +0.8]$ points) across every JEPA variant.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Live streaming is much harder than retrospective review. Comparing Panels (a) and (b) of Table IV shows why 1D latent scores degrade under causal reporting: buffered attribution averages residuals across every overlapping window covering a timestamp, whereas causal streaming scores a single right-edge step with no averaging, allowing one or two drifting coordinates to inflate the shared SPOT threshold (Section I). On these development streams, a rate-matched Bernoulli alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$ hits 28.1% of events at $\leq 1.3\%$ causal FPR, versus 60.0% at TimesNet's 24.3%.

C. Exploratory Horizon Sweep Across $S \in \{1, 4, 16, 32, 64\}$

Shorter horizons trade buffered smoothing for lower live lag ($S-1$ steps). Retraining on five exploratory streams (Fig. 3(a); at $S = 1$ all prefixes coincide), both panels exceed 1D Multi-Horizon SPOT at every S with test FPR $\leq 3.6\%$. At $S = 1$ and $S = 4$, *Laminar-JEPA* reaches $2.6\times$ and $4.3\times$ scalar causal Point-F1 ($0.037, 0.050$ vs. $0.014, 0.012$; 77%–84% ER) and *Laminar-JEPA-Lite* reaches $2.1\times$ and $3.1\times$ ($0.030, 0.036$; 64%–82% ER), cutting delay at $S = 4$ to 36.7–44.1 steps, whereas buffered Point-F1 peaks at 0.290 ($S = 32$) and 0.255 ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot set a general rule.

D. Design-Stage Operating Budgets and Fault Taxonomies

Fig. 3(b) and Table V examine the 19 development streams across budgets and fault types: (1) *Diagnostic budgets (Fig. 3(b), Table II(c)):* At a post-hoc matched $f^* = 1\%$ test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD; this ranking advantage over scalar JEPAs does not carry over to the archive (Section VII-E). (2) *Contextual vs. point faults (Table V):* Out-of-range spikes ($> 5\sigma$) are caught by reconstructors and amplitude rules alike, whereas in-range contextual faults alter cross-channel phase or velocity within normal bounds; on NASA contextual sequences [4], *Laminar-JEPA-Lite* and *Laminar-JEPA* reach 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR versus 62.5% at 41.5% FPR for TimesNet.

E. Primary Held-Out Generalization, False-Alarm Tails, and Calibration Limits

Tables VI and VII and Fig. 3(c) report the disjoint five-block archive. We label each result by its operating point: *autonomous* (deployed healthy-only thresholds), *matched calibration rate* (every detector set to the same healthy alarm rate), *post hoc* (thresholds matched on test labels, diagnostic only), and *buffered* (retrospective review).

1) *Live event coverage and delay (autonomous, Table VI(a)):* Across the 118 untouched streams, *Laminar-JEPA-Lite* raises causal ER from 52.5% to 63.8% ($+11.3$ points, cluster CI $[+5.1, +21.8]$, $9/9$ datasets, sign-flip $p = 0.0039$) and nearly halves detection delay (70.8 vs. 133.5 steps). Pooled causal Point-F1 moves only from 0.076 to 0.079; the stream-level $p = 0.014$ coexists with stream and cluster CIs that include zero ($[-0.019, +0.023]$ and $[-0.033, +0.060]$), so we treat this gain as modest and sensitive to the inference procedure. The gain depends on calibration length. With $N_{\text{cal}} \geq 500$ ($n = 59$), causal Point-F1 doubles ($0.040 \rightarrow 0.079$, CI $[+0.016, +0.064]$, $p = 2.7 \times 10^{-5}$; $0.042 \rightarrow 0.077$ in the independent rerun, $p = 3.7 \times 10^{-5}$) with $+16.0$ ER points, and stays ahead without Exathlon ($n = 43$: 0.051 vs. 0.031, $p = 0.0050$), at a higher alarm burden on healthy test steps (median FPR_{test} 5.5% vs. 0.6%). With $N_{\text{cal}} < 500$, the scalar baseline has higher causal Point-F1 (0.112 vs. 0.080, $p = 0.31$) although the panel catches more events (75.0% vs. 68.4%). Under *buffered* review, ER rises from 59.0% to 67.4% and Point-F1 from 0.143 to 0.156 ($p = 0.022$) at lower buffered FPR_{hold} ($19.5\% \rightarrow 14.3\%$).
2) *Where the live gain comes from (matched calibration rate and post hoc, Tables VI(b), VII(b)):* At the panel's own low calibration rate (median 0.25%), it catches more causal events than the scalar baseline (65.0% vs. 52.6%, $p = 1.7 \times 10^{-5}$) and than Fisher, top-k, and max-z pooling

TABLE IV
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR_{hold} ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR_{hold} ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR_{hold} ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table II(a,b); best in **bold**, second underlined. Δt = mean step delay over detected events; DS-F1 = unweighted 7-dataset macro Buffered Point-F1. In (c), n = streams, K = channels, E = events, π = faulty-step fraction, $F1_{all} = 2\pi/(1+\pi)$ (trivial always-alarm Point-F1), Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

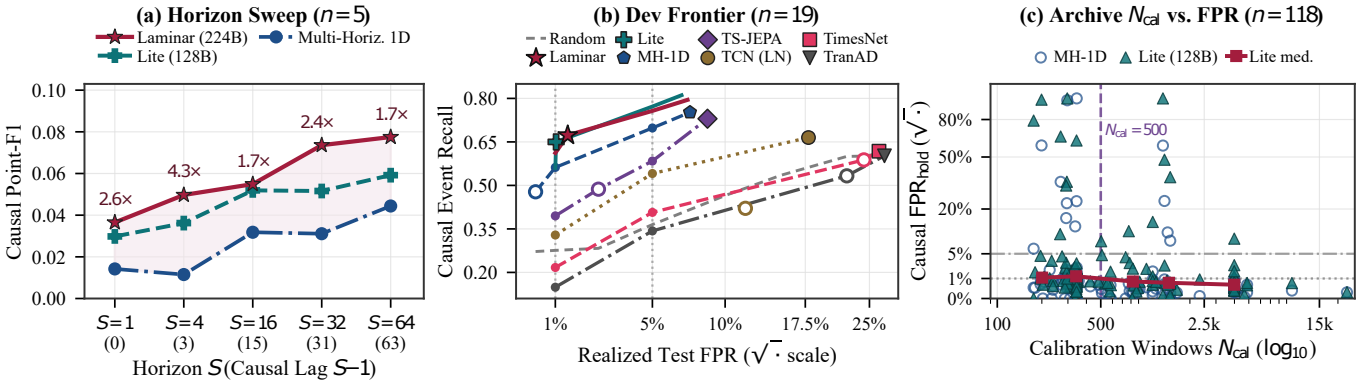


Fig. 3. Horizon scaling, operating frontier, and archive calibration-size effects. **(a)** Exploratory horizon sweep ($S \in \{1..64\}$, causal lag $S-1$, n=5 streams, nominal FPR $\leq$ 3.6% vs. 2.5% $\rightarrow$ 21.3% for TimesNet). **(b)** Design-stage causal ER vs. realized test FPR ($\sqrt{\cdot}$ scale, n=19 streams) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted), autonomous points, and $q^* \leq 0.05$. **(c)** Primary archive calibration windows N_{cal} vs. causal held-out FPR_{hold} across n=118 untouched streams (Spearman $\rho = -0.32$, $p = 5.2 \times 10^{-4}$), marking N_{cal}=500 and 1%/5% budgets.

of the same residuals (61.1%, 58.1%, 58.8%). At common healthy alarm rates of 0.5%–5% these differences shrink: the panel leads Fisher by 1–3.5 ER points (significant at 2% and 5%) while Fisher has equal or higher Point-F1, top-k pooling is nearly indistinguishable, max-z matches its ER, and the scalar baseline trails by 3–6 ER points at statistically similar Point-F1. At post hoc matched test FPR the panel shows no ranking advantage: its ER ties the scalar baseline and trails max-z, and the scalar baseline has higher Point-F1 at $f = 1\%$ (0.084 vs. 0.049, $p = 0.024$). The archive gains of the deployed panel therefore come mainly from its label-free operating point, a quorum that settles at higher recall and shorter delay than scalar SPOT, rather than from a better continuous ranking. On these residuals the quorum is one competitive coordinate aggregator, not a uniformly superior one.

3) *False-alarm tail (autonomous, Table VII(a), Fig. 3(c)):* For *Laminar-JEPA-Lite*, causal FPR_{hold} has mean 7.0%, median 0.78%, 90th percentile 10.6%, and 95th percentile 50.2%; 15, 13, 10, and 10 streams exceed 5%, 10%, 20%, and 30% (13, 11, 8, and 5 for Multi-Horizon 1D), and 54.2% and 87.3% of streams meet 1% and 5% budgets. The ten worst streams carry 79% of all false-alarm mass. Short calibration contributes (Spearman $\rho = -0.32$ between N_{cal} and FPR_{hold}; mean 16.0% for $N_{cal} < 250$ vs. 1.05% for $N_{cal} \geq 2{,}500$) but is not the only cause: three of the ten worst streams have $N_{cal} > 1{,}300$. The strongest predictor is a shift between the healthy calibration and hold blocks, measured by the KS distance between their quorum counts ($\rho = 0.63$; median D_{KS} 0.55 above 10% FPR vs. 0.16 at or below 5%, Mann–Whitney $p = 1.6 \times 10^{-6}$), whereas N_{eff} is uncorrelated with FPR_{hold} ($\rho = -0.01$). Eight streams with $D_{KS} \geq 0.5$ carry 66% of the panel's and 69% of the scalar baseline's false-alarm mass, so these healthy

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**ER**	**FPR$_{\text{hold}}$**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=.05	f^*=.01	f^*=.05	q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Buf. PR = buffered PR-AUC; ER and FPR$_{\text{hold}}$ = autonomous causal recall and held-out FPR; q^*=.05 = 95th-percentile $\mathbf{X}_{\text{cal}}$ ER; f^*=.01, .05 = diagnostic FPR$_{\text{test}}$-matched ER; best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 [2100, 2210]; point: A-1, P-3, D-3, D-14, D-15, C-1 [550, 750]). (b) n=19 non-GHL streams split into in-range ($\max|z| \leq 5\sigma$) vs. out-of-range ($\max|z| > 5\sigma$) causal events.

TABLE VI
DISJOINT FIVE-BLOCK UNTOUCHED ARCHIVE BENCHMARK (n=118)

	Pt-F1 ↑		ER ↑		FPR$_{\text{hold}}$		Caus. FPR Tail			
(a) Archive (n=118)	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Med.**	**p90**	**p95**	≤5%
TS-JEPA (1D)	.057	.134	.414	.459	**.042**	**.110**	**.004**	.070	**.209**	88.1
Multi-Horiz. 1D	.076	.143	.525	.590	.044	.195	**.004**	**.068**	.237	**89.0**
Laminar-JEPA (224B)	.072	.144	.609	.619	.068§	.190	.013	.076	.383	86.4
Laminar-Lite (128B)	**.079**†	**.156**‡	**.638**¶	**.674**	.070	.143	.008	.106	.502	87.3
Adequate (N_{cal}≥500, n=59) vs. short (N_{cal}<500, n=59) histories										
MH-1D, N_{cal}≥500	.040	.096	.368	.457	**.020**	.133	**.002**	**.009**	**.086**	**93.2**
Lite, N_{cal}≥500	**.079**	**.127**	**.528**	**.577**	.047	**.099**	.006	.084	.169	88.1
MH-1D, N_{cal}<500	**.112**	**.191**	.684	.727	**.069**	.258	**.007**	**.175**	**.364**	84.7
Lite, N_{cal}<500	.080	.186	**.750**	**.773**	.093	**.188**	.012	.323	.669	**86.4**

	Causal						Buffered		
(b) Rerun (n=118)	**Pt-F1**	**ER**	**ER$_{.05}$**	**Mean**	**Med.**	**Δt**	**Pt-F1**	**ER**	**ER$_{.05}$**
Multi-Horizon 1D (SPOT)	.074	.502	.752	**.046**	**.003**	136	.145	.586	.422
at Lite's cal. rate	**.085**	.526	.752	.050	.005	110	.146	.576	.422
Laminar-JEPA-Lite	.078	**.650**	.726	.071	.010	**66**	.154	.661	**.533**
at 1D's cal. rate	.061	.563	.726	.062	.004	95	**.157**	**.698**	**.533**
Laminar-JEPA, 224 dials	.074	.646	**.767**	.068	.013	70	.143	.643	.498
Laminar-JEPA, 160 dials	.073	.616	.736	.064	.009	88	.140	.634	.529

(a) Three seeds; best **bold**, second underlined: MH-1D = Multi-Horizon JEPA (1D), Lite = Laminar-JEPA-Lite; ≤5% = % of streams with causal FPR$_{\text{hold}}$≤5%. PR-AUC (caus./buf.): TS-JEPA .155/.207, 1D .162/.230, 224B .143/.211, Lite .145/.224; unquenched 224B: .075/.141 F1, .644/.640 ER, .069/.190 FPR$_{\text{hold}}$. †Lite vs. 1D causal F1 Δ=+0.003 (stream CI [−0.019, +0.023], cluster CI [−0.033, +0.060]; 70/43 wins, p=0.014); vs. TS-JEPA +0.022 (cluster CI [+0.005, +0.052]). ‡Buf. F1 Δ=+0.014 (p=0.022). ¶Causal ER Δ=+0.113 (cluster CI [+0.051, +0.218], 9/9 sets) at higher causal FPR$_{\text{hold}}$ (+0.025, cluster CI [+0.006, +0.055]). §Disjoint quench: FPR$_{\text{hold}}$ Δ=−0.0007 (p=0.55), ER −0.035. (b) Independent retraining on identical splits, used only for operating-point controls (headline values come from (a)); Mean/Med. = causal FPR$_{\text{hold}}$; ER$_{.05}$ = diagnostic recall at 5% test FPR; "at cal. rate" matches $\mathbf{X}_{\text{cal}}$ alarm rates (Fisher .077 F1, .611 ER; top-k .069, .581; max-z .063, .588; Stouffer .057, .558; ℓ_2^2 conf. .071, .556; BH-FDR .024, .146).

regime changes defeat both; outside them, the panel's mean causal FPR$_{\text{hold}}$ is 2.6%. Healthy test steps show the same drift (median/mean FPR$_{\text{test}}$ 3.0%/15.2% vs. 0.9%/8.8% for 1D).

4) *Quantile reliability and deployment conditions:* The quorum count has median $\tau_{\text{int}} = 7.5$ and median $N_{\text{eff}} = 65$, so its 99.5% quantile has fewer than one expected effective exceedance ($N_{\text{eff}} < 200$) on 105/118 streams; we call a level q *supported* only when $N_{\text{eff}}(1-q) \geq 1$. Replacing the quorum quantile by its distribution-free 90% upper confidence bound with N_{eff} samples (calibration maximum when unsupported) lowers mean causal FPR$_{\text{hold}}$ only from 7.1% to 5.8% (95th percentile 50% → 42%) while cutting ER from 65.3% to 47.7% and Point-F1 from 0.080 to 0.050: conservative quantiles do not remove alarms caused by regime shifts. Autonomous use therefore requires three conditions: (i) $N_{\text{cal}} \geq 500$, below which the scalar baseline may be preferable (choosing the panel only when $N_{\text{cal}} \geq 500$ raises pooled causal Point-F1 to 0.095); (ii) an acceptance check in which the frozen detector stays within the alarm budget (e.g., 5%) on an untouched healthy block such as $\mathbf{X}_{\text{hold}}$; and (iii) drift monitoring with recalibration after regime changes. On the 52 streams meeting (i) and (ii), the panel stays ahead (causal Point-F1 0.074 vs. 0.038, $p = 3.0\times10^{-4}$) at mean FPR$_{\text{hold}}$ = 0.9%, yet mean FPR$_{\text{test}}$ is 12.4%, so (iii) remains necessary; other streams need operator supervision. On short histories ($T_{\text{nom}} < 2{,}500$, $n = 18$), two-fold cross-fitting cuts causal FPR$_{\text{test}}$ from 15.4% to 3.8% at 69.9% ER.

5) *Regime drift vs. tail misfit:* On development streams, chronological versus block-permuted threshold refits attribute 71.3% (TS-JEPA) and 78.3%–81.8% (reconstructors) of excess causal FPR to inter-block drift, versus 18.7% with centering, without ruling out tail misfit.

TABLE VII
ARCHIVE FALSE-ALARM AND OPERATING-POINT AUDIT (n=118, CAUSAL)

				FPR$_{\text{hold}}$		FPR$_{\text{test}}$
(a) Highest-FPR streams	N_{cal}	N_{eff}	D_{KS}	**Lite**	**1D**	**Lite**
Exathlon 6_1_500000_65	1315	110	1.00	1.00	.47	1.00
SMAP D-11	297	12	1.00	1.00	.97	1.00
Exathlon 10_2_1000000_67	199	14	1.00	.99	.74	1.00
Daphnet S09R01E4	176	77	.96	.86	.06	.90
SMAP G-7	268	48	.96	.63	.45	.31
SMAP R-1	343	343	.62	.62	.10	1.00
Exathlon 5_5_1000000_91	1347	803	.48	.48	.31	.62
Exathlon 10_3_1000000_75	1473	53	.75	.37	.10	.99
SMAP D-8	295	295	.34	.34	.23	.14
SMAP D-9	292	292	.32	.32	.55	.06

(b) Matched budgets	**Lite**	**Fisher**	**Top-k**	**Max-z**	**1D**	**FPR$_{\text{hold}}$**
r=0.5% (healthy)	.676/.084	.667/**.090**	.666/.084	.662/.072	.620/.096	.06–.09
r=1%	.732/.096	.715/**.099**	.711/.096	.738/.086	.703/.109	.07–.10
r=2%	.791/.112	.761/.112	.785/.111	.819/.103	.751/.120	.08–.12
r=5%	.863/.131	.828/.133	.861/.130	.873/.124	.823/.135	.13–.17
f=1% (post hoc)	.529/.049	.525/**.054**	.533/**.049**	**.570**/.052	.549/**.084**	–
f=5%	.723/.099	.719/**.107**	.727/.100	**.785**/.099	.754/.122	–

Independent three-seed rerun, five-block protocol, seed-averaged per stream. (a) The ten highest Lite causal FPR$_{\text{hold}}$ streams (the same ten as in Table VI(a)); $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}}$ of the Lite quorum count; D_{KS} = KS distance between $\mathbf{X}_{\text{cal}}$ and $\mathbf{X}_{\text{hold}}$ counts. (b) Causal ER/Point-F1 when every detector is thresholded at the same healthy $\mathbf{X}_{\text{cal}}$ alarm rate r (label-free) or post hoc at the same test FPR f (uses test labels; diagnostic). Fisher, top-k, and max-z pool the same frozen Lite residuals; 1D = Multi-Horizon JEPA (1D). Bold/underlined: significantly higher/lower than Lite (paired Wilcoxon p<0.05). Last column: range of mean FPR$_{\text{hold}}$ across the five detectors.

VIII. CONCLUSION AND FUTURE WORK

We studied multi-horizon circuit-breaker panels (*Laminar-JEPA-Lite* for operational deployment; 224-dial *Laminar-JEPA* as an ablation reference) for abrupt and intermediate-onset telemetry faults across 33 exploratory design-stage streams and 118 primary untouched archive streams:

1) *Event coverage and delay:* Under the five-block archive protocol, *Laminar-JEPA-Lite* raises live causal ER from 52.5% to 63.8% on all nine datasets ($p = 0.0039$) and nearly halves detection delay, with 35%–52% fewer parameters and 2.43× faster inference than *Laminar-JEPA*. Pooled causal Point-F1 rises only modestly (0.076 → 0.079) but doubles when $N_{\text{cal}} \geq 500$ (0.040 → 0.079); on shorter histories the scalar baseline is preferable on Point-F1. Weight quenching gives no reliable benefit.
2) *Operating point:* At matched healthy alarm rates the panel performs on par with simpler coordinate pooling, and at matched test FPR it does not rank faults better than the scalar baseline. Its value lies in a label-free quorum, fitted to correlated trip counts, that settles at a more useful recall and delay than scalar SPOT.

3) *False-alarm tail and deployment:* The panel's causal $\mathrm{FPR}_{\mathrm{hold}}$ has mean 7.0%, median 0.78%, 90th percentile 10.6%, and 95th percentile 50.2% (Table VI(a)). Eight streams with healthy regime shifts carry two thirds of all false alarms for both detectors, so autonomous use requires $N_{\mathrm{cal}} \geq 500$, a held-out acceptance check, and drift monitoring.
4) *Limits:* An optional channel branch helps synthetic single-sensor faults only at a higher FPR, and slow ramps ($L \geq 4C$, including GHL at 0.000 Point-F1) and natural sensor failures remain open.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.
[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.
[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.
[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.
[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.
[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.
[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.
[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.
[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.
[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.
[11] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.
[12] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
[13] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.
[14] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.
[15] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.
[16] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.
[17] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.
[18] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.
[19] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.
[20] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.
[21] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.
[22] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.
[23] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.
[24] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.
[25] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.
[26] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
[27] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.
[28] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.
[29] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.
[30] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.
[31] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.
[32] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.
[33] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.
[34] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.
[35] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.
[36] H. Hotelling, "Multivariate quality control, illustrated by the air testing of sample bombsights," in *Techniques of Statistical Analysis*. McGraw-Hill, 1947, pp. 111–184.
[37] C. A. Lowry, W. H. Woodall, C. W. Champ, and S. E. Rigdon, "A multivariate exponentially weighted moving average control chart," *Technometrics*, vol. 34, no. 1, pp. 46–53, 1992.
[38] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.
[39] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.
[40] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.
[41] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.
[42] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.
[43] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.
[44] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.
[45] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.
[46] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.
[47] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.
[48] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.
[49] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.
[50] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.
[51] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.
[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.